DECODE 2025-2026

ENGINEERING PORTFOLIO

14023

SPRINGBANK ROBOTICS TEAM

PHOENIX RESURRECTION

THE TEAM

Liam
Team Lead
Build/Design

Stupid Sexy Jones
Co gn

Hunter
Team Lead
Drive/Build

Ethan
Drive/Build

Laurana
Code/Build
Design

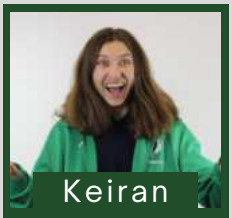
Arianna
Portfolio
Code/Finace

Fiona
Drive/Build

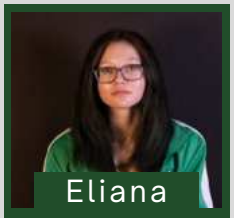
Carter
Drive/Finace

Riley
Senior Code

Simon
Junior Build

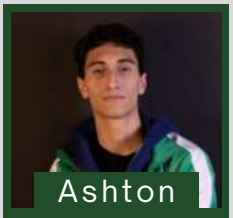


Keiran
Code



Eliana
Build

Angil
Build



Ashton
Code

Charlie
Code

Harrison
Build

Phoenix
Mascot

Maddison
Robot

Meghan
Robot

MISSION STATEMENT

> We **inspire** innovation in robotics, empowering our team and community through STEM education and collaboration, fostering future leaders in technology.

The Springbank Robotics Team (SRT) Phoenix Resurrection, 14023, is a 9th-year FTC team in Calgary, Alberta.
Our **student-led** team works together to learn, grow and collaborate together, with our experienced members mentoring and helping out our newcomers. We **connect** to our community through outreach, determined to inspire future generations through STEM and robotics.

CONNECT : We explore ways to achieve our goals and engage with the local STEM community!

Our Goals	Achieving these Goals:
Achieve Worlds	As a team, we have decided to push our skills to improve and become the **best versions of ourselves**. By excelling beyond our previous performances at competition, we are able to track our progress in improvement.
Engage in STEM outreach	We aim to **inspire and support** younger students and our local community by sharing our passion for STEM. This includes: • Volunteering for youth robotic programs • Participating and hosting in community events • Offering hands-on opportunities that encourage curiosity and learning
Give back to FIRST	We are **grateful** for the opportunities FIRST provides to **inspire others in STEM**. To give back, our team **volunteers** at FIRST events and competitions, helping with setup, and assisting teams. By contributing our time, we help **strengthen the community** that has supported our own growth

MENTORS

Mr. Mathew Mentor

Ms. Heidi Mentor

Our FTC robotics team is lucky to have amazing mentors like Mr. Mathew and Ms. Heidi, who guide us through everything from building and programming to strategy and teamwork.

Our mentors are very important, but at its core, SRT is a **student-led team**. Being student-led helps us build skills like leadership, problem-solving, teamwork, and creativity while learning from our mistakes. Students in grades 9–12 **support and mentor each other,** and every member has a role regardless of experience. With mentors guiding us and students leading the work, we maintain a strong balance between support and independence.

CREDITS

This Engineering Portfolio was created by Team 14023, using Canva and Google Docs. We also referenced the TWCA FTC Engineering Portfolio Guide and used the FTC-recommended WebAIM Contrast Checker.

SOCIAL MEDIA

This season we have even more people involved in **social media**, specifically our **Instagram**. Having social media is a great way for our team to **connect** with other teams, showcase everything to parents, other students and our whole community. It is a great way to **promote** our team and the **achievements** we are making as well. Additionally, social media is one of the ways our **sponsors can stay updated** on the team.

As well as Instagram, we stay **connected** with our **community by the SRT newsletter**. We have our own newsletter with the latest schedules, updates and more, and we include this newsletter in our schools newsletter as well, the **Phoenix Flyer**.

SPRINGBANK ROBOTICS TEAM (SRT)

November 2025 Newsletter

Gearing Up for Competition Season!

The Springbank Robotics Team is off to a strong start this year! With two competitive teams — 14021 and 14023 — the lab has been buzzing with building, coding, redesigning, and testing. Both teams are progressing extremely well and are on track to be fully prepared for our first competition of the season. Great job to all students on the teamwork, creativity, and dedication so far!

Fundraising Update

Fundraising remains critically important to our program. This year we are working toward hosting a major Community of Springbank Raffle, which requires SRT to supply $4,000 worth of prizes.

$4000
$3000
$2000

ALUMNI INVOLVEMENT

Many of our **SRT alumni** stay **involved** with FIRST and come back to visit our team. We have had many of our **alumni**, even those who graduated years ago, come back this year and check in and see how the team is doing. We have even had alumni come to the competitions! Our alumni are a great reminder of our **team's environment** and the **impact** that stays with each student.

TEAM MEETINGS

We hold **regular SRT team meetings** that include our team, 14023, sister teams, and our junior members. Our goal is to **involve everyone in all aspects** of the robot, team sustainability, fundraisers, and outreach. In our latest meeting, we debriefed on our recent competition (provincial qualifiers), discussing successes, areas for improvement, and **gathering everyone's input**.

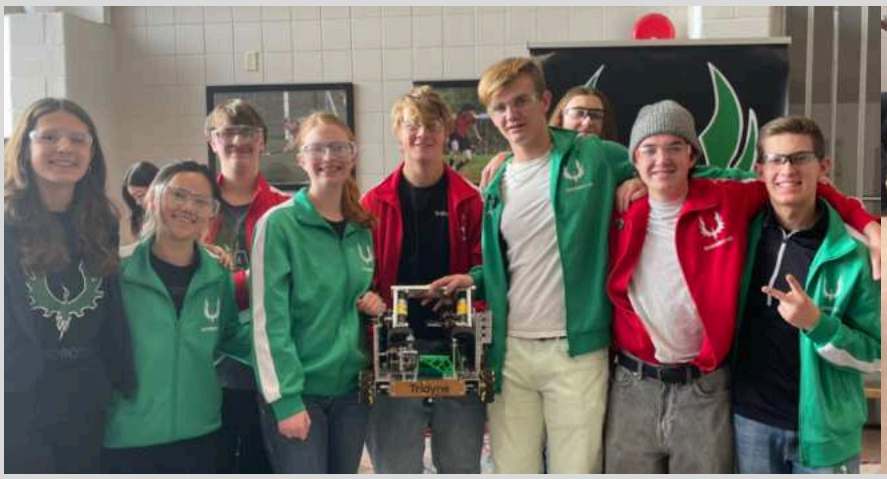
REACH: Our efforts to inspire others to embrace the FIRST culture, while introducing them to the exciting world of STEM.

OUTREACH

Outreach plays an **essential** role in promoting FTC and **motivating** the next generation of innovators. By building connections with schools, local businesses, and the community, SRT highlights the **excitement of robotics** and the wide range of opportunities within STEM (science, technology, engineering, and mathematics). These efforts help strengthen community support, develop leadership skills within our team, and encourage others to get involved. Through outreach, our team continues to **grow** while advancing the broader goal of preparing tomorrow's engineers, scientists, and problem-solvers.

TOTAL OUTREACH HOURS

DATE	EVENT	HOURS	IMPACT
1/20/2025 to 1/23/2025	Supporting Coding week at Elbow Valley Elementary School	18	200+ people
2/4/2025	SHAD Ocean Presentation	1	40 people
3/4/2025	Family and Friends Event at SCHS - Annual Event, 2026 TBD	2.5	60 people
11/12/2025	Telus Spark Scrimmage	4	50+ people
1/19/2026 to 1/22/2026	Supporting Coding week at Elbow Valley Elementary School	20	200+ people
And much more to come in the future!			



HIGHTLIGHTS FOR THIS SEASON

Our Grade 9s took the opportunity to **volunteer** for FIRST at our first meet (Clay Pot Meet #2). Their enthusiasm showed **Gracious Professionalism** as they worked hard to assist others by supporting teams in the pit area and helping matches run smoothly.

We attended the Telus Spark Scrimmage hosted by FTC, where our team had the opportunity to showcase our robot and **demonstrate innovation** to young, curious students. Through hands-on interaction and demonstrations, we **inspired** interest in robotics and STEM while sharing our passion with the next generation.

ADAPTING TO THE STRIKE

Due to the Alberta teachers' strike, our team temporarily moved to a member's house to continue work on the FTC robot. This quick change kept us on track with **design, build, and programming**. The obstacle highlighted our ability to stay **focused** despite the challenges, strengthening our **commitment** to **perseverance** and **teamwork**, values that continue to inspire us throughout the season.

INSPIRING FUTURE GENERATIONS

Every year, our team also takes the opportunity to visit Elbow Elementary School **to inspire the younger students**. We taught them the basics of coding, creating, innovating, and CAD, giving them hands-on exposure to STEM concepts. Through this outreach, we aimed to **encourage curiosity, creativity**, and a lasting **interest** in **engineering and technology**.

SUSTAINABILITY:

Our team's long-term success relies on sustainable funding and financial support.

FUNDRAISING EFFORTS

Fundraising is a key part of our success, providing the resources needed to **cover the costs** of building our robots and attending competitions. These efforts allow us to focus on the technical and strategic aspects of the team while ensuring we have the materials and opportunities to compete at our best. Every effort we make, and every contribution we receive, directly supports our ability to compete, innovate, and grow as a team, making fundraising a vital part of both our **present success and future goals**

SPONSOR SUPPORT

We constantly work to provide meaningful benefits to **our sponsors** and maximize the value of their support. We invite sponsors to visit our classroom and competitions, and we also visit their offices to see how they work, and apply it to our robot design process. We stay **connected** through regular updates so sponsors can see the direct impact of their support. In return for their guidance and financial support, we **proudly promote them** on our robot, banners, social media, and other team materials, while continuously brainstorming **new ways to give back**

FINANCIAL SUPPORT TIMELINE:

10/2025 - 12/2025
Purdy's Holiday Fundraiser
+$2366.44

12/2025
Tridyne Projects
+$2500

1/19/2026
Team Bottle drive
+$495.05

1/2026 - 3/2026
Springbank 50/50 Raffle
+$10 000 (Expected)

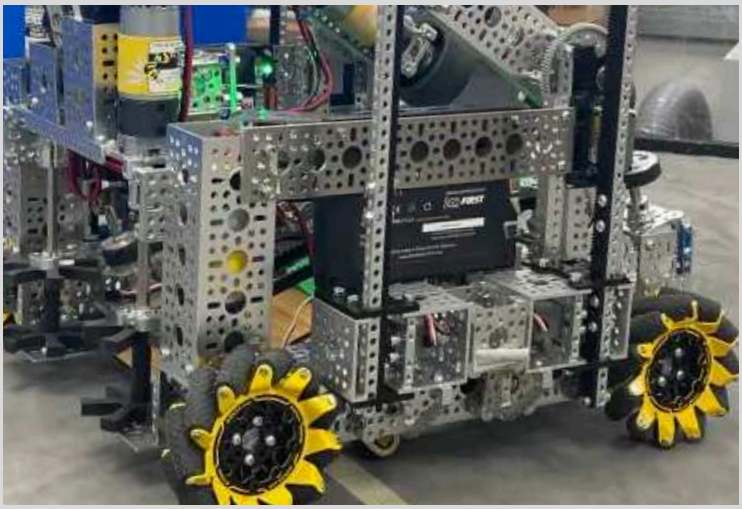
...AND MANY MORE!

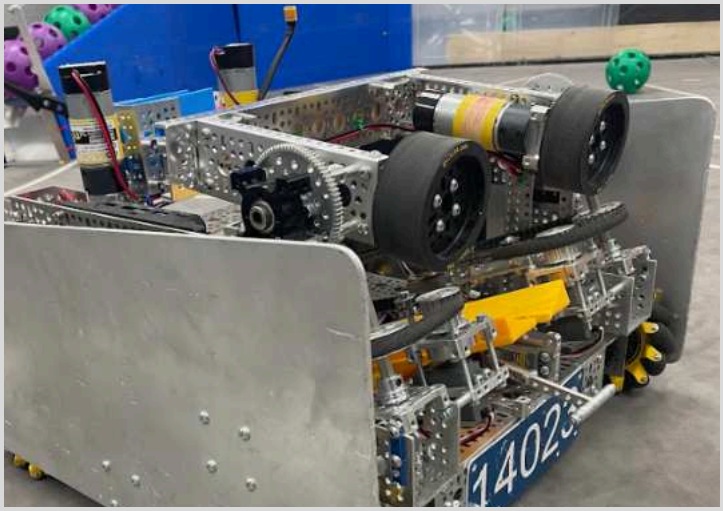
DESIGN:

Meet the robots of our team, explore our robot iterations and discover our unique build ideas.

MADDISON'S OVERVIEW/STRATEGY

Maddison, named in honor of an SRT alumni, features an **active intake** with a **dual conveyor system** to efficiently collect artifacts and deliver them to the flywheels for consistent scoring. Our robot also includes an **adjustable flywheel launch system** and **mecanum wheels**, allowing for quick maneuvers and versatile strategies. Given the high-contact nature of this season's games, we designed **rigid aluminum side panels that can tank impacts**, allowing our drivers to stay focused on scoring.





ROBOT DESIGN FEATURES

Drivetrain

- 435 RPM motors powering mecanum strafe wheels from goBuilda, allowing us to achieve holonomic movement.

Intake

- Two 1620 RPM motors with GoBuilda boot wheels, intaking artifacts without pushing them away.

Conveyor Belt

- Two symmetrical super-speed motors integrated into a pulley system with GoBuilda timing belts to move the artifacts from intake to shooters

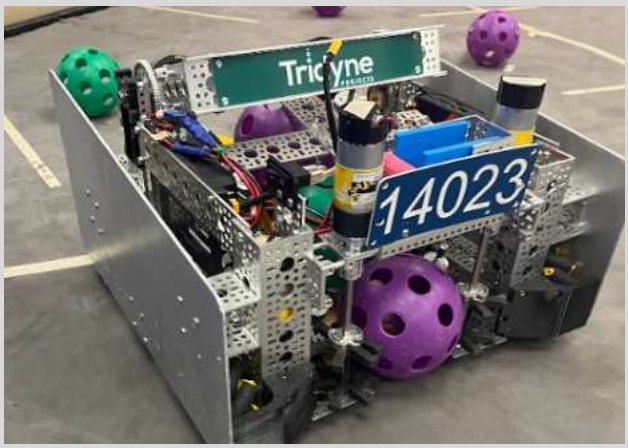
Shooter

- Modified 6000 RPM motors with adjustable angles and Rhino wheels, enabling shots from anywhere on the field.

THINK: Introducing our design proccess, robot evolution, and how we improve ourselves through creative thinking

1. Brainstorm:
As a group, we discuss ways to solve problems with our robot, collaborating to develop and refine plans for improvement.

2.Sketch:
We then drew a rough draft of what the robot would look like on a whiteboard so that we could envision our creation.



3. Design & CAD:
After sketching our solutions, we begin creating prototypes and models of our ideas using **Autodesk Fusion360** to make sure parts would fit properly and would work efficiently. This process involved close collaboration within the team, ensuring everyone's input is included!

4. **Assembly:**
Once all parts have been designed or ordered our build team. Using the 3D-printed parts combined with successful prototypes.

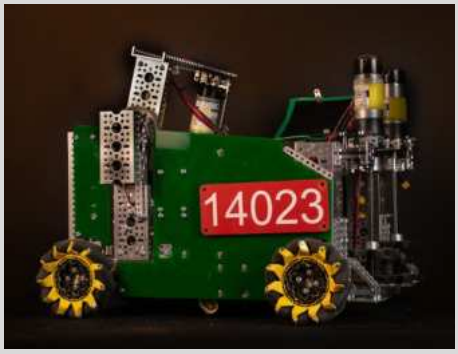
5. Evaluate & Iterate:
Once assembled, we **evaluate** how the parts interact with the robot, ensuring that all parts function correctly and reliably. We test the robot, evaluate its performance, and refine it through **iterations**, improving reliability and consistency. Finally leading to the completed robot and revised CAD.

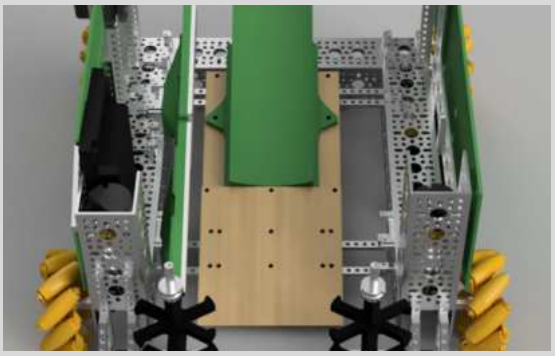
THEN WE REPEAT THIS PROCESS!

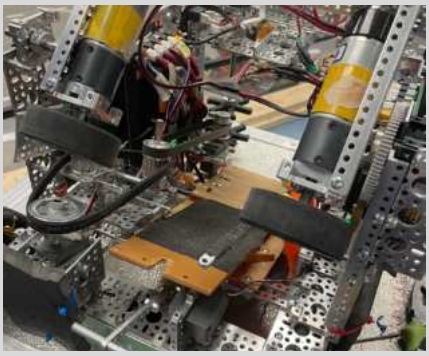
ROBOT EVOLUTION - CLAYPOT MEET TWO (AFTER EVENT #1)

MEGHAN, OUR FIRST ROBOT ITERATION

Our initial robot design was named **Meghan**, after a fellow alumni. Meghan featured a **one-sided conveyor belt system** that moved artifacts from the intake to a set of **adjustable gecko-wheel shooters**. To accommodate potential partners with lifts in endgame, we designed the robot to be **small and compact**, which also enabled quick maneuvers and greater adaptability on the field.



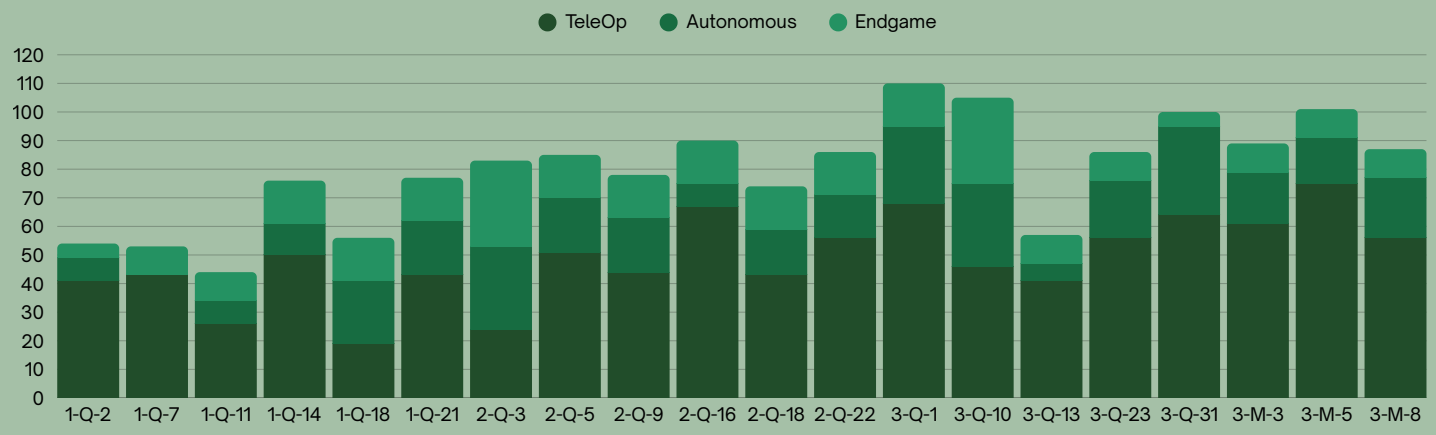




RETIRING TO A BACKUP ROBOT

After multiple competitions with **Meghan,** we decided to create an entirely new robot chassis inspired by our previously successful design. This led to our current robot: **Maddison**. As a result, Meghan has been retired to a backup role, allowing us to **experiment,** fine-tune code, and let us prototype new ideas - all without **interrupting** work on our new robot. Meghan has also been able to serve as a **catalyst** for junior members' **development** in their technical skills and driving abilities. Thanks to Meghan, the future of our program is looking bright.

MATCH PERFORMANCE GRAPH THROUGHOUT THE SEASON

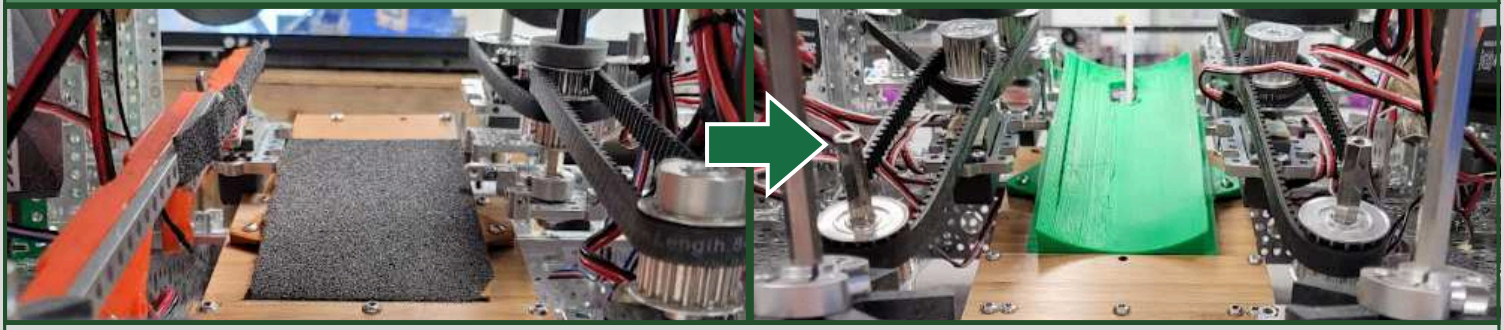


This graph displays our performance per game in our matches. This showcases our **improvement** throughout the season

ROBOT EVOLUTION - ARROWHEAD MEET TWO (AFTER EVENT #2)

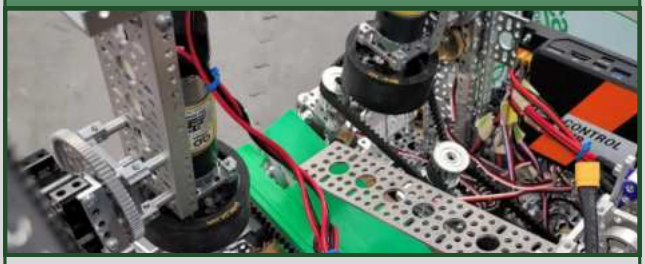
After this event, we transitioned to our second robot, Maddison (see Overview for details). This page highlights the key improvements made from Meghan's to Maddison's design.

CONVEYOR BELT CHANGES



The robot uses a **3D-printed angled ramp** to move artifacts from intake to shooter via a conveyor. Our first design had a **single sided belt**, causing artifacts to slip back down. We improved it by using **wider goBILDA timing belts** and **adding a second belt** on the opposite side, gripping artifacts from both sides for reduced slippage and a more consistent feed into the shooter.

FLY WHEELS SWAP



We noticed our shots **drifting left and right** because the **Gecko flywheels** were **expanding and compressing** while spinning, causing inconsistent artifact launches. We fixed this by switching to **Rhino wheels**, which **do not expand or compress**, resulting in more consistent shooting in both driver practice and autonomous routines.

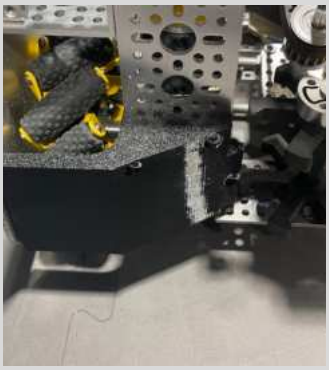
GENERAL DESIGN CHANGES

Maddison has a **symmetrical** body for **precise autonomous performance** and a wider U-channel for **better maintenance access**. This improved on Meghan's design, which had tedious wiring and adjustments.

To prevent artifacts from **contacting the flywheels early**, we added a **servo** with a block at the end of the ramp, letting drivers focus on driving and reducing potential penalties.

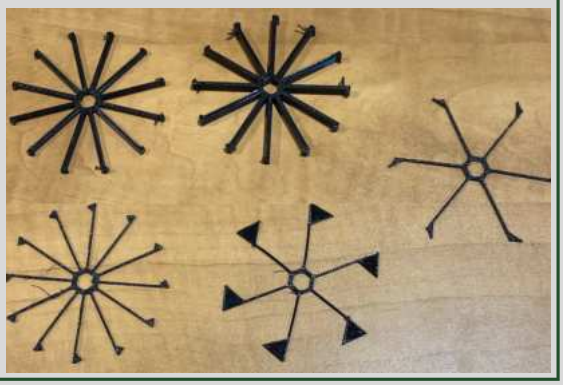
ROBOT EVOLUTION - NAA'MIIS'STII INTERLEAGUE TOURNAMENT (AFTER EVENT #3)

WHEEL BUMPER/FUNNEL AND TPU INTAKE

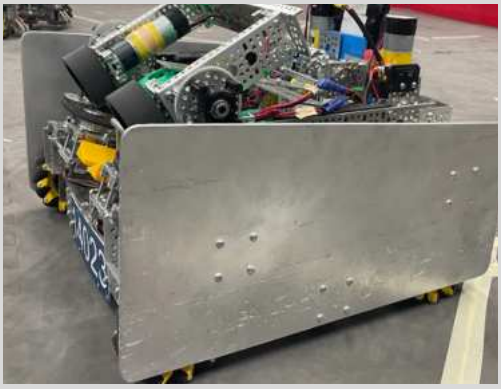


Throughout our matches, we realized we were struggling to **intake artifacts stuck in the corners** of the field. To address this, we designed a **unique 3D-printed wheel bumper** that also **funnels artifacts** directly into our intake. This allows us to **reach into corners** we previously couldn't, while **protecting our wheels** from damage.

When we couldn't intake artifacts from the corners, we **designed and 3D printed** new intake wheels. Although it worked, it didn't significantly improve performance or fully solve the issue, so we chose not to use it. Instead, using the bumpers with a funnel proved far **more effective**.



ALUMINUM SIDE PANELS



Using a **CNC plasma cutter**, we designed and manufactured **custom aluminum side panels**. Choosing metal instead of lighter materials **intentionally increased** the robot's weight, allowing **more resistance** to **defensive contact** and helps maintain **stable positioning** for **consistent performance.**

DRIVER PRACTICE AND MOCK MATCHES

In our group debrief, we identified a short amount of **driver practice time** as a key weakness. To address this, we **increased** practice sessions and **organized mock matches with junior robots** and our former robot **Meghan**.

This helped our drivers refine their skills while giving juniors a chance to showcase their robots they worked hard on. (See page 13 for more details).

INNOVATE:

Meet our prototype designs, junior robots, and experimented ideas!

SCRAPPED LIFT SYSTEM

Our lift system uses **four servos with a gear-and-pinion setup** on the side panels, allowing teammates to park beneath us. While it provides **valuable endgame points**, the lift is risky as side impacts can lead to a potential fall from 18 inches. In competition, we found that the lift system added **significant weight**, which **reduced** autonomous speed and consistency and led to lost autonomous points. Additionally, despite our thin side panels providing increased clearance, most teams were still **unable to fit underneath**, making the lift rarely usable. Due to its low effectiveness, added risk, and negative impact on overall performance, we decided to **remove** the system for provincial.

JUNIOR PROTOTYPES

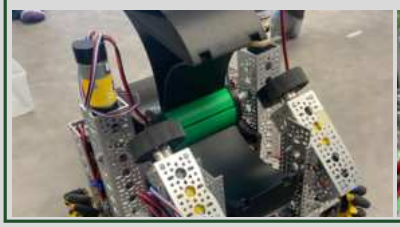
As a **student-led team**, our senior members mentor juniors by working with them on non-competitive robots built from spare parts. This **low-pressure environment** allows junior members to learn freely and gain **hands-on experience** without risk.

PROTOTYPE LIFT SYSTEM

One example is a **prototype lift system** designed by younger members using **REV bars and dual Viper Slides**, evenly distributing weight and enabling **significant lift height.**

PROTOTYPE 3D PINWHEEL

Another prototype featuring a **3D-printed pinwheel system** that funneled artifacts into a rotating mechanism for **efficient organization**. Combined with computer vision, it could automatically sort pieces into the **correct pattern to maximize points**.



CONTROL:

We use software and hardware in various ways to improve our robots functionality during matches!

AUTONOMOUS CODE

For our autonomous, we used Java with **Roadrunner** with **three dead wheels**.
Our strategy emphasized adaptability, with **12 different autos** enabling us to shoot from **multiple positions**, **coordinate** with alliance partners, **minimize collisions**, and **maximize points**

```
// ------------------------
@Override
public void runOpMode() throws InterruptedException {

    // init shooter
    Spinner spinner = new Spinner(hardwareMap);
    Shooter shooter = new Shooter(hardwareMap);
    Conveyor conveyor = new Conveyor(hardwareMap);
    ShooterAngle shooterAngle = new ShooterAngle(hardwareMap);

    // start pose
    Pose2d beginPose = new Pose2d(new Vector2d( x: 0,  y: 0),  heading: 0);

    // build drive
    MecanumDrive drive = new MecanumDrive(hardwareMap, beginPose);

    // build trajectories
```

ODOMETRY PODS

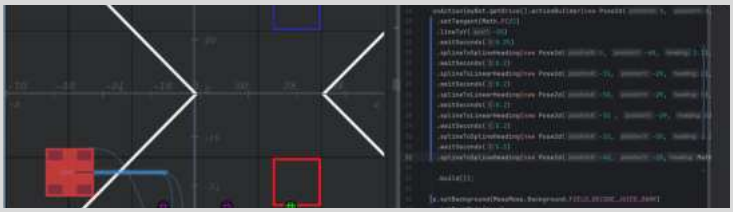
For our autonomous, we use **dead wheels**, unpowered omni wheels equipped with rotary encoders to track the distance traveled. This data is processed through a **kinematic equation** and integrated to determine the robot's relative position on the field.
The advantage of dead wheels over drive wheel encoders is their **minimal slip**, especially when compared to mecanum wheels. This significantly enhances **accuracy**, particularly during **high acceleration**, making it ideal for our mecanum drive setup. With this system, we can fine-tune our autonomous code for **optimal performance.**

PROBLEMS	PROGRAM SOLUTION
Collecting the artifacts was sometimes **inconsistent** due to **different acceleration** between trajectories, which **reduced reliability** during collection as the robot would sometimes knock them out of line.	This was solved by adding controlled pauses and slowing approach speeds, allowing the robot to **stabilize** and **align** more closely with objects before intake. While this reduced overall speed, it **significantly improved consistency** and **reliability** during **autonomous** collection of artifacts.
Due to the **significant weight** of our robot, which was necessary for **stability and durability**, its autonomous performance was considerably **slower**, making it significantly more difficult to achieve a 9-artifact autonomous.	By increasing our RoadRunner tuning's **maximum acceleration** and adjusting kV (Volts*Velocity), kA (Volts*Acceleration), and kS (Volts), we **boosted our speed**. Through **trial and error**, and a few blown battery fuses, we found a sweet spot that didn't overload our motors but was fast enough for a **9-artifact autonomous.**

TESTING AUTONOMOUS CODE LIBRARIES

ROADRUNNER

Road Runner is a **coordinate-based** motion library developed by ACME Robotics team for FTC competitions. It enables our robot to execute **complex, high-velocity, high-acceleration trajectories** with advanced path-following capabilities, ensuring **precise and reliable movement**.



PEDROPATHING

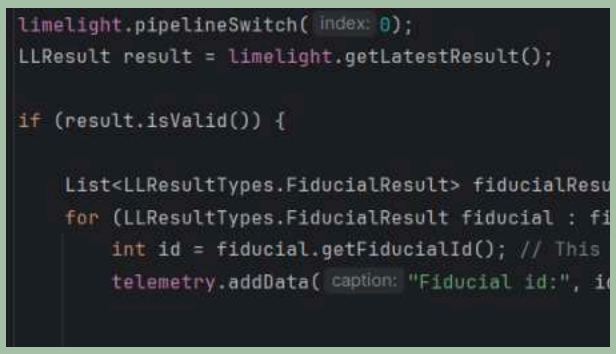
Designed by Team 10158, PedroPathing is an advanced **Reactive Vector Follower** that adapts to obstacles in **real time**. Though harder to set up, it's faster, **more accurate**, and improves **autonomous reliability** over traditional RoadRunner.

We have been working with both Java Libraries, comparing and contrasting their strengths and limitations.
PedroPathing adjusts in real time during a path, making it responsive and precise, but it is highly battery-dependent and can be volatile. In contrast, RoadRunner only adjusts after autonomous, reducing real-time accuracy. However, it remains reliable under voltage drops, an important advantage this year as we run multiple motors at full power

CAMERA INTEGRATION

Our team has also been experimenting with **Limelight Smart Cameras**, which are specialized for robotics and April Tag detection. Although not yet mounted on the robot, we've been testing it in a non-competitive setting to explore features like **pattern recognition** to improve autonomy.

Our use of Limelight in practice was **successful**, but **impractical** for our current design. Implementing this idea took to long. The working code was able to **consistently detect the obelisk april-tag**. This told the robot which trajectories to follow for **accurate and precise autonomous**.



Additionally, we are learning **OpenCV**, a computer vision tool for **real-time processing and machine learning**. These preparations will support **smooth future integration** to further enhance our **robot's performance**.

TELE-OP DRIVE CODE

Our robot uses the built-in OnBot Java in the REV Hardware Client. Our code is **robot-centric**, and our compact design allows for faster movement, improved stability, and efficient endgame performance.

Our tele-op code includes preset motor positions for **different shooting angles**, making it adaptable from various locations on the field. Additionally, **adjustable motor speeds control shooting power**, allowing us to account for factors such as defensive robots and field positioning without overshooting. Lastly, We also implemented **multiple drive speeds** to ensure accurate parking during endgame.

```
aitForStart();
f (isStopRequested()) return;

hile (opModeIsActive()) {

    // MOVEMENT INPUT
    double y = -gamepad1.left_stick_y;
    double x = gamepad1.left_stick_x * 1.1;
    double rx = gamepad1.right_stick_x * turn

    // Strafe correction
    double strafeCorrection;
    if (x > 0) strafeCorrection = -Math.pow(x
    else if (x < 0) strafeCorrection = Math.p
    else strafeCorrection = 0;

    rx += strafeCorrection;

    // SPEED MODES
    if (gamepad1.right_trigger > 0) { DriveSp
    if (gamepad1.b) { DriveSpeed = 0.5; turns
    if (gamepad1.left_trigger > 0) { DriveSpe


    // WHEEL POWER CALC
    double denominator = Math.max(Math.abs(y)
    double fl = (y + x + rx) / denominator *
    double bl = (y - x + rx) / denominator *
```

PROBLEMS	PROGRAM SOLUTIONS
During tele-op, we ran out of buttons on our controllers, and the **existing button system** was confusing, making it hard for drivers to control the robot	To solve this, we added a **toggle button** for the **conveyor belts** and **intake motors**. This freed up buttons on the controller and made it easier for drivers to **quickly turn off the intake system**. The toggle button **switches between the on and off state** with each press. By changing the **boolean value** we allow the single button to operate both in the on and off positions.
Our robot would drift during tele-op. Unfortunately, this issue was **difficult to fully resolve.** The above problem was mechanical and would throw off weight distribution.	We programmed a **strafe correction** feature into our tele-op drive code. This correction **adjusted** for the **uneven motor outputs by automatically compensating for drift.** This helped the robot maintain a straight path and allowed our drivers to **navigate** the field smoothly and accurately.

THANK YOU FOR REVIEWING OUR PORTFOLIO, AND FOR YOUR SUPPORT! -PHOENIX RESURRECTION, 14023